



What is Cluster Analysis Transcript

Hello everyone, welcome to lesson 1 video 2. In this video, we will study hierarchical clustering method. This method builds clusters step by step, either by merging smaller groups or by splitting larger ones. Now that we have learnt a lot of statistic and statistical concepts, we will try making them useful or how do we use them in various different types of algorithms.

And one specific type of algorithm which groups data together or which tries to find groups within historical data is known as cluster analysis. So just to give you an example what exactly is cluster analysis, let's say you have a bag of or a basket of mixed fruit in which you have apples, guavas, bananas and everything. Now because you know which are apples, what is their colour, what is their shape and various different properties, you group them all together.

Now let's say you segregate them as in you take away all the apples and form a different group, take away all the bananas, form a different group, take away all the guavas and form a different group. So you have created a group of similar fruits. Now in this particular type of categorisation or clustering, you are basically forming natural groups which exist in the data such that your data might be arranged in a random way and you are trying to find out are there patterns in that random data.

With that let's begin and let's explore more about cluster analysis. So cluster analysis is actually an algorithm which is a part of a broader category which is called as unsupervised learning and think of how do we learn, how humans learn. I'll tell you about examples how we make machines learn which we eventually call as machine learning but the learning can be of two different types or actually they are of three different types.

One is supervised, supervised as in you learn under the supervision of someone and that is with a teacher. Let's say when you are a kid, you learn under the supervision of your parents, teachers and so on right and they tell you that this is apple, this is guava, this is banana and you tend to recognise that apple is red in colour and it has a round shape and so on and orange has a yellow colour sorry orange colour and round shape and so on right. So you tend to find different specifics and you see them and you are also told about what class they belong to or basically what types they are, what are apples, what are oranges, what are bananas and so on.

Now because you are told that is supervision, sometimes you are not told about what is there in the data okay and then what you generally try to do is to find common patterns or similarities or some patterns in the data which could help us in grouping data points. Now in this part of learning which is called as unsupervised, there comes a lot of algorithms. We are only specifically looking into clustering or cluster analysis because clustering is one of the most widely used algorithm and is one of the simplest as well so that you understand it, you are able to apply it, you are able to recognise the importance and everything.

So the analogy here is instead of being told which photons contains a cat or which photos contains a cat, the algorithm is given a thousand of photos and asked to find out groups of photos that look similar. Let me give you another example. Let's say you have a lot of photos on Google photos or your phone, whatever software it uses.

If you would want to club together all the photos coming from a particular individual wherein there is one particular individual, let's say you want to club together all the photos wherein you are there. So that clubbing of all the photos where you are there and clubbing of all the photos where your mother is there or your brother is there and so on, that exactly is clustering. Clustering is grouping similar things together.

So let's say I have photos of mountains, I can cluster them together. I have photos of seas, I can cluster them together. So clustering similar thing together or grouping similar thing together is what clustering is.

So cluster is a grouping method, a way to find out groups in our unorganised or random data. So here in this example, we are seeing data before or unorganised data, random data, which is here and we have to organise it, let's say, or we have to club it in clusters. Now after applying clustering, what could happen is because these three data points if you see on the left, they look similar or nearby to each other.

One of the ways of finding whether they are similar is within whatever dimension your data has, high dimensional data to find out are the data points looking closer to each other, right? So you can use distances. Because these points are closer, you can say that these form a one single cluster and that's why you see that they are labelled as or they are tagged as blue. Now similarly here, you see these four, they are quite near to each other, hence they are tagged as green.

Similarly that's what happens to these three here on the right hand side. But there is one data point which is left alone. There are no groups which are near to it, hence this might be tagged as an outlier.

That's what happens in clustering. You have a random data, historical data on which you apply clustering algorithm and then you group together similar items based on similarities and one of the methods of using similarity is distances. Now the key principle of clustering is to actually identify which data points are near and which data points are far away or which groups are near, which groups are far away, right? So you would want to have high intra-cluster similarity, okay? Data points within the same cluster should be all similar, homogeneous and low inter cluster similarity.

Data points in different clusters should be as dissimilar as possible, okay? So within the cluster, the data points should be similar while between the clusters, the data points should be dissimilar. And this is because if you see here, if you have this particular cluster, they are one tagged into one cluster because they are similar to each other while this particular cluster, they are similar to each other and that's why they are



second group. When you compare these two clusters, you see that between clusters, the objects are different, right? So that's what the key principle is.

Between the clusters, you need to find more and more homogeneity. Between the clusters, you need to find more variance or you need to have more dissimilarities and within the clusters, you need to find more homogeneity, right? Now where is this used? This is used very widely in NBFCs, bankings for doing marketing or let's say upselling. Let's say someone has bought or has loans from a particular bank and the bank employees would want to give you credit cards, they might see that which particular group you are similar to and because you are similar to a particular group which has got loans or credit cards, they would recommend you to buy loans because you have much more similarity to the kind of group which is already having credit cards or loans, right? And then in biology, you could group genes with similar expression patterns to identify functional relationships, right? Similarly, you could club together all the documents which are similar like every topic of politics or sports or business, they can be categorised.

They are also used for anomaly detection. Let's say there is an anomaly in the high dimensional data and you would want to find out whether there are not so similar values or some outliers which are present in the data just like the example you saw here, right? So this is not so similar data point. So you might want to find out anomalies in the data and for that also you use clustering.

These are basic common applications wherein it is used. I hope this beginning lecture was good for you to understand what exactly is clustering. In the next lecture, we will dig deeper into what other types of things do we need to know in clustering, okay? In some way, hierarchical clustering provides a clear and interpretable structure for grouping data.

The dendrogram allows us to visualise how clusters form at each and every stage. In the next video, we will explore K-means clustering, one of the most powerful clustering methods that we use in industry today.

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